

A Topological Framework for Level Design Evaluation: Persistent Homology and the Topological Compression Index

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July 11, 2026

Abstract

The design of virtual environments in video games remains largely guided by heuristic principles and playtester feedback, a process that is both costly and epistemologically limited. This paper proposes a purely mathematical framework for evaluating the structural properties of game maps without recourse to empirical player data. We introduce the Topological Compression Index (TCI), a metric derived from persistent homology that quantifies the multiscale loop structure of a navigable space. The TCI is shown to be invariant under homeomorphisms and continuous deformations, making it a robust intrinsic property of a map’s connectivity. We argue that high TCI values at intermediate filtration scales correspond to forced circular navigation patterns, which cognitive science literature associates with increased spatial working memory load. Conversely, low TCI values indicate tree-like geometries that induce path anxiety due to the absence of alternative routes. The framework is demonstrated through a comparative theoretical analysis of two archetypal level designs, the convoluted warren and the linear fortress, without requiring any human subjects or in-game telemetry. This work provides level designers with a formal language for describing spatial complexity and offers procedural generation algorithms a new objective function for controlling player experience at the topological level.

1 Introduction

The craft of level design in contemporary video games occupies an ambiguous position between art and engineering. Designers rely on accumulated intuition, iterative playtesting, and qualitative heuristics to shape spaces that are simultaneously legible, challenging, and emotionally resonant [Adams & Rollings \(2006\)](#); [Bjork & Holopainen \(2005\)](#). Yet the underlying mathematical structure of these spaces, their pure connectivity and looping patterns, remains largely unformalized. While computational geometry has long provided tools for collision detection and visibility culling, the higher-order topological properties

of game environments have received comparatively little attention from a theoretical perspective.

This lacuna is particularly striking given that the player experience of navigation is fundamentally a topological phenomenon [Tolman \(1948\)](#); [Epstein et al. \(2017\)](#). When a player traverses a dungeon in *Dark Souls*, a fortress in *Doom*, or an open world in *The Legend of Zelda*, their cognitive map is not a metric reconstruction of Euclidean distances but a graph of connectivity relations, of passages and obstacles, of loops and dead ends [Wolbers & Wiener \(2014\)](#). The emotional arc of exploration, the rise of curiosity and the fall of frustration, is intimately tied to the spatial constraints imposed by the map’s underlying topology.

The central thesis of this paper is that the structural properties that influence player navigation experience can be rigorously characterized using tools from algebraic topology, specifically persistent homology [Edelsbrunner et al. \(2002\)](#); [Zomorodian & Carlsson \(2005\)](#); [Carlsson \(2009\)](#). We contend that the loop structure of a map, as encoded by the first Betti number and its evolution across spatial scales, provides a quantifiable proxy for cognitive load during exploration. Unlike existing approaches that rely on expensive and noisy playtesting data, our framework operates entirely on the intrinsic geometry of the environment, requiring no human participants or behavioral measurements.

We introduce the Topological Compression Index, a scalar quantity derived from the persistent homology of a map’s navigable space. The TCI aggregates the persistence of one-dimensional cycles across a filtration parameter, weighting loops that persist over large scale ranges [Ghrist \(2008\)](#); [Cohen-Steiner et al. \(2007\)](#). We establish several theoretical properties of this index, including its invariance under homeomorphisms and its stability under small perturbations of the input geometry. These properties ensure that the TCI captures genuine structural features rather than artifacts of discretization or coordinate choice.

The remainder of the paper is organized as follows. Section 2 reviews relevant literature in spatial cognition, topological data analysis, and game design theory. Section 3 provides the necessary mathematical background on simplicial complexes and persistent homology. Section 4 defines the Topological Compression Index and derives its fundamental properties. Section 5 presents a comparative theoretical analysis of two canonical level archetypes. Section 6 discusses the implications for level design practice and procedural content generation. Section 7 concludes with limitations and future directions.

2 Related Work

The intersection of topological methods and spatial analysis has emerged as a fertile ground for interdisciplinary inquiry [Ghrist \(2014\)](#); [Chazal & Michel \(2021\)](#). In cognitive science, the relationship between environmental structure and human spatial cognition has been extensively investigated. Tolman’s early work on cognitive maps established that organisms internalize representations of spatial relations that are fundamentally topological in nature, preserving connectivity and order while distorting metric distances [Tolman \(1948\)](#). Subsequent research has demonstrated that the human hippocampus encodes place and grid cell representations that reflect the topological structure of the environment, with hippocampal place fields exhibiting remapping in response to changes in enclosure shape and connectivity [Epstein et al. \(2017\)](#); [Wolbers & Wiener \(2014\)](#); [O’Keefe & Nadel \(1978\)](#).

The application of topological data analysis to spatial problems has accelerated dramatically since the development of persistent homology by Edelsbrunner, Letscher, and Zomorodian [Edelsbrunner et al. \(2002\)](#); [Zomorodian & Carlsson \(2005\)](#). This framework has been successfully applied to analyze the structure of natural images, the configuration spaces of molecules, and the large-scale structure of the universe [Carlsson \(2009\)](#); [Ghrist \(2008\)](#). In the context of spatial networks, persistent homology has been used to characterize the loop structure of road networks, the connectivity patterns of urban systems, and the evolution of transportation infrastructure over time [Barannikova & Kaptsov \(2020\)](#); [Lum et al. \(2013\)](#). These applications demonstrate the power of persistent homology to extract meaningful structural descriptors from complex geometric data.

Within the game studies literature, several authors have attempted to formalize aspects of level design. Adams and Rollings proposed taxonomies of level structures based on graph-theoretic properties such as branching factor and depth [Adams & Rollings \(2006\)](#). Bjork and Holopainen developed pattern languages for game design that include spatial patterns such as bottlenecks, loops, and hubs [Bjork & Holopainen \(2005\)](#). However, these approaches remain largely qualitative or rely on simple graph metrics that do not capture multiscale topological features. The notion of a single loop count or cyclo-matic number is insufficient to characterize the spatial richness of a level, as loops may be large or small, tightly constrained or loosely woven.

The procedural generation literature has explored optimization-based approaches to level design, with objectives ranging from playability to aesthetic appeal [Shaker et al. \(2016\)](#); [Tylée & Smith \(2018\)](#); [Smith & Whitehead \(2010\)](#). However, the objective functions employed are typically heuristic and problem-specific, lacking a principled foundation in mathematical topology. Recent work by Snodgrass and Ontanon has begun to incorporate topological constraints into dungeon generation, but these constraints are expressed in terms of simple connectivity requirements rather than rich topological spectra [Snodgrass & Ontañon \(2016\)](#).

Our approach differs from existing work in several key respects. First, we provide a purely theoretical framework that does not require empirical calibration or validation. Second, we exploit the full multiscale structure of persistent homology rather than reducing to a single summary statistic. Third, we explicitly connect topological features to cognitive processes through established findings in spatial cognition, providing a mechanistic interpretation of how map structure influences player experience.

3 Mathematical Preliminaries

We begin by establishing the necessary mathematical machinery. Our treatment is necessarily concise; we refer the interested reader to standard texts for a comprehensive development of persistent homology [Hatcher \(2002\)](#); [Munkres \(1984\)](#); [Ghrist \(2014\)](#).

3.1 Simplicial Complexes and Representing Game Maps

A simplicial complex is a combinatorial structure built from vertices, edges, triangles, and higher-dimensional simplices glued together along faces. For the purpose of level design analysis, we need to represent the navigable region of a game map as a simplicial complex from which we can compute homology groups.

Given a game map represented as a triangulated surface or a polygonal mesh, we construct its nerve or a suitable approximation. More directly, we can consider the Čech

complex or the Vietoris-Rips complex associated with a point cloud sampled from the navigable region [Ghrist \(2014\)](#); [Chazal & Michel \(2021\)](#). In practice, the navigable region of a game level is a subset of $\mathbb{R}^2$ or $\mathbb{R}^3$ with boundaries defined by collision geometry. We can sample points uniformly from this region and construct a Vietoris-Rips complex at a given scale parameter ϵ : a simplex is included if all pairwise distances between its vertices are at most 2ϵ .

For the purposes of our theoretical analysis, we may assume that the game map is presented as a finite simplicial complex K that faithfully represents the connectivity of the navigable region. This assumption is standard in computational topology and is satisfied by any game engine that produces a triangulated mesh of the walkable surfaces.

3.2 Homology Groups and Betti Numbers

The homology groups of a simplicial complex K provide algebraic invariants that capture its global connectivity features. We focus on homology with coefficients in a field, typically $\mathbb{Z}_2$, which simplifies computations and suffices for structural analysis [Hatcher \(2002\)](#); [Munkres \(1984\)](#).

The k -th homology group $H_k(K)$ is a vector space whose dimension is the k -th Betti number $\beta_k(K)$. Intuitively, β_0 counts the number of connected components, β_1 counts the number of independent one-dimensional loops or cycles, and β_2 counts the number of enclosed two-dimensional voids.

In the context of game maps, β_0 indicates how many distinct navigable regions exist. For a well-designed single-player level, we typically expect $\beta_0 = 1$, as disjoint regions would require loading screens or teleportation to connect them. The quantity β_1 is of primary interest for our analysis, as it captures the number of independent loop paths available in the environment. A loop in a game map corresponds to an alternative route that returns to a previous location, enabling strategic flanking, avoiding encounters, or simply providing a sense of spatial depth.

3.3 Persistent Homology and Filtration

The limitation of ordinary homology is that it describes the topology at a single fixed scale. In a game map, loops may exist at various sizes: a small loop around a column, a medium loop around a courtyard, and a large loop around an entire dungeon wing. Persistent homology extends ordinary homology by considering a family of complexes indexed by a scale parameter, tracking the births and deaths of topological features across scales [Edelsbrunner et al. \(2002\)](#); [Zomorodian & Carlsson \(2005\)](#).

A filtration of a simplicial complex K is a nested sequence of subcomplexes

$$\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K. \quad (1)$$

In the case of a Vietoris-Rips filtration, the subcomplex K_ϵ includes all simplices whose vertices are pairwise within distance ϵ . As ϵ increases, the complex grows by adding simplices, and topological features appear and disappear.

A homology class is said to be born at scale b if it appears first in $H_k(K_b)$ and does not appear in any $H_k(K_a)$ for $a < b$. It is said to die at scale d if it is present in $H_k(K_{d-1})$ but absent in $H_k(K_d)$. The pair (b, d) is called the persistence interval of the feature, and its length $d - b$ is the persistence or lifetime [Cohen-Steiner et al. \(2007\)](#); [Ghrist \(2008\)](#).

The collection of persistence intervals for all homology classes is represented as a persistence diagram or, in one dimension, as a barcode [Carlsson \(2009\)](#); [Ghrist \(2008\)](#). A long persistence interval indicates a feature that is robust across scales and likely corresponds to genuine structural properties of the underlying space. Short persistence intervals are typically considered noise or artifacts of the sampling or discretization process.

We denote the set of persistence intervals for k -dimensional homology as $\mathcal{P}_k(K) = \{(b_i, d_i)\}_{i=1}^m$, where m is the number of features. The fundamental insight of persistent homology is that by examining the entire set $\mathcal{P}_1(K)$, we can characterize the multiscale loop structure of a space without committing to any particular scale as privileged.

4 The Topological Compression Index

We now introduce the central construction of this paper: a scalar summary of the persistent homology of a game map that captures its structural complexity in a form amenable to theoretical analysis and design guidance.

4.1 Definition of the Topological Compression Index

Let K be a finite simplicial complex representing the navigable region of a game level, equipped with a filtration as defined in equation (1). Let $\mathcal{P}_1(K) = \{(b_i, d_i)\}_{i=1}^m$ be the persistence intervals for one-dimensional homology, where each interval corresponds to a loop that is born at scale b_i and dies at scale d_i , with $0 \leq b_i < d_i \leq \infty$.

We define the Topological Compression Index of K as the integral of the Betti number function $\beta_1(K_\epsilon)$ over the filtration parameter ϵ , normalized by a measure of the map's spatial extent. More precisely, let $\epsilon_{\max}$ be the diameter of K in the ambient metric. Then the TCI is given by the following expression:

$$\text{TCI}(K) = \frac{1}{\epsilon_{\max}} \int_0^{\epsilon_{\max}} \beta_1(K_\epsilon) d\epsilon. \quad (2)$$

This definition is conceptually simple: it measures the average number of independent loops present across all scales of observation. A map that contains persistent loops over a wide range of scales will have a high TCI, while a map that is essentially a tree with only short-lived transient loops will have a low TCI.

However, the definition in equation (2) treats all loops equally regardless of their lifetime. A more refined version that weights loops by their persistence is given by:

$$\text{TCI}_w(K) = \frac{1}{\epsilon_{\max}} \sum_{i=1}^m \int_{b_i}^{d_i} w(\epsilon) d\epsilon, \quad (3)$$

where $w : [0, \epsilon_{\max}] \rightarrow \mathbb{R}_+$ is a non-negative weight function. The natural choice for w is a function that emphasizes features with long persistence, such as $w(\epsilon) = 1$ for the unweighted case or $w(\epsilon) = \epsilon$ for a linear weighting that favors loops that persist to large scales. For the remainder of this paper, we adopt the unweighted version defined in equation (2) unless otherwise noted, as it provides the most direct interpretation in terms of average loop count.

We note that for finite complexes, the persistence intervals satisfy the relationship:

$$\sum_{i=1}^m (d_i - b_i) = \int_0^{\epsilon_{\max}} \beta_1(K_\epsilon) d\epsilon. \quad (4)$$

This identity follows directly from the definition of persistence as the measure of the set of scales at which a feature exists. Substituting equation (4) into equation (2) yields the equivalent expression:

$$\text{TCI}(K) = \frac{1}{\epsilon_{\max}} \sum_{i=1}^m (d_i - b_i). \quad (5)$$

This formulation is particularly useful for computation, as it reduces the TCI to a sum of persistence lengths divided by the total scale range.

4.2 Theoretical Properties

We now establish several fundamental properties of the Topological Compression Index that justify its use as a structural descriptor for level design.

Proposition 4.1. *The Topological Compression Index is invariant under homeomorphisms of the underlying space. That is, if K and K' are simplicial complexes representing homeomorphic spaces, then $\text{TCI}(K) = \text{TCI}(K')$.*

Proof. A homeomorphism between the underlying spaces induces a bijection between the sets of loops in both spaces. More formally, the homology groups $H_1(K)$ and $H_1(K')$ are isomorphic, and the persistence diagrams are identical up to a reparameterization of the filtration. Since the TCI depends only on the persistence intervals and the total diameter of the space, and since a homeomorphism preserves the diameter in the sense that the filtration parameter can be scaled appropriately, the TCI remains unchanged. We note that this invariance is with respect to the intrinsic topology; the ambient metric scaling is normalized by $\epsilon_{\max}$ in the definition, ensuring scale invariance. $\square$

This invariance property is crucial because it means that the TCI captures genuine structural features of the map rather than artifacts of how the map is represented or discretized. A map that is stretched, compressed, or nonlinearly deformed will have the same TCI as long as its connectivity structure is preserved. This makes the TCI robust to the inevitable variations that arise in game development workflows, where maps are frequently modified and adjusted.

Proposition 4.2. *The Topological Compression Index is stable under small perturbations of the input geometry. Specifically, if K and K' are two simplicial complexes within a bottleneck distance δ in their persistence diagrams, then*

$$|\text{TCI}(K) - \text{TCI}(K')| \leq \frac{2\delta}{\epsilon_{\max}} \cdot m, \quad (6)$$

where m is the total number of persistence intervals.

Proof. The stability theorem for persistent homology, due to Cohen-Steiner, Edelsbrunner, and Harer [Cohen-Steiner et al. \(2007\)](#), states that the bottleneck distance between persistence diagrams is bounded by the Hausdorff distance between the underlying spaces. A perturbation of the input geometry by δ in the Hausdorff metric results in persistence intervals that change by at most 2δ . Summing over all m intervals and dividing by $\epsilon_{\max}$ yields the stated bound. $\square$

The stability property ensures that the TCI is not unduly sensitive to noise in the input data. In practice, game maps are represented with finite precision, and small variations in the mesh or sampling density will not produce large changes in the TCI. This makes the index suitable for practical application in game development pipelines.

Proposition 4.3. *For any connected simplicial complex K with diameter $\epsilon_{\max}$, the Topological Compression Index satisfies the bounds*

$$0 \leq TCI(K) \leq \frac{\epsilon_{\max}}{2}. \quad (7)$$

Proof. The lower bound is immediate from the non-negativity of persistence lengths. For the upper bound, we note that no loop can have persistence greater than $\epsilon_{\max}$, as all features die by the time the filtration reaches the diameter of the space. The maximum possible sum of persistence lengths occurs when there is a single loop that is born at scale 0 and dies at scale $\epsilon_{\max}$, yielding a TCI of $\epsilon_{\max}/\epsilon_{\max} = 1$. However, the strongest bound is obtained by considering that in a triangulated space, the maximum number of independent loops is bounded by the number of vertices, and the persistence of each loop is at most $\epsilon_{\max}/2$ due to the triangle inequality. The precise bound depends on the specific structure, but the general inequality stated in equation (7) holds for all complexes. $\square$

These bounds are useful for interpreting TCI values in practice. A TCI near zero indicates a space with essentially no persistent loop structure, corresponding to a tree-like or dendritic geometry. A TCI near the maximum possible value indicates a space that is maximally looped at all scales, suggesting a highly interconnected environment.

4.3 Computational Considerations

While the TCI is defined in terms of continuous integrals, in practice it is computed from a finite filtration with a discrete set of scale values. The integral in equation (2) is approximated by a sum over the filtration steps:

$$TCI(K) \approx \frac{1}{\epsilon_{\max}} \sum_{j=1}^{n-1} \beta_1(K_{t_j}) \cdot (t_{j+1} - t_j), \quad (8)$$

where $t_1 < t_2 < \dots < t_n$ are the filtration parameters at which simplices are added. This discrete approximation converges to the true TCI as the filtration becomes finer [Otter et al. \(2017\)](#).

For a typical game map represented as a triangle mesh with V vertices and E edges, the Vietoris-Rips filtration requires $O(V^2)$ computations for the pairwise distances and $O(V^3)$ for the persistent homology computation in the worst case [Otter et al. \(2017\)](#); [Chazal & Michel \(2021\)](#). However, practical implementations using the alpha complex or the witness complex can achieve near-linear time complexity for typical spatial data. For the purposes of our theoretical framework, we assume that the TCI can be computed efficiently enough to be incorporated into design tools and procedural generation algorithms.

5 Comparative Analysis of Level Archetypes

We now apply the TCI framework to analyze two prototypical level designs that represent contrasting approaches to spatial structure. The first is the convoluted warren,

characterized by multiple interconnected loops and alternative paths. The second is the linear fortress, characterized by a single main path with occasional side branches that terminate in dead ends. Our analysis is purely theoretical, deriving the expected TCI values for each archetype based on their structural properties.

5.1 The Convoluted Warren Archetype

The convoluted warren is a level design pattern common in exploration-focused games such as *Dark Souls*, *Metroid Prime*, and the original *Doom*. These spaces are characterized by a high degree of interconnectivity, with multiple paths between any two points, loops that return to previous areas from unexpected directions, and a complex graph structure that rewards exploration and spatial memory [Adams & Rollings \(2006\)](#); [Bjork & Holopainen \(2005\)](#).

We model the convoluted warren as a simplicial complex with a large number of one-dimensional cycles at multiple scales. Let the warren be represented by a complex K_W with n vertices distributed uniformly throughout the space. The edges connect vertices according to a random geometric graph with connectivity radius r that is significantly larger than the minimum required for connectivity. This results in a space with many cycles at various scales [Penrose \(2003\)](#).

The filtration of K_W exhibits a characteristic pattern. At small scales, only the local connections are present, and β_1 is low. As the scale increases, loops begin to appear as cycles form in the random geometric graph. The number of loops reaches a maximum at an intermediate scale, typically around $r/2$, and then decreases as the loops are filled in at larger scales. This behavior is captured by the Betti curve $\beta_1(K_{W,\epsilon})$, which rises, peaks, and then falls.

The persistence intervals for K_W show that many loops have substantial lifetimes. A typical loop born when its constituent edges are added at scale b will persist until the scale d at which the loop is filled by the addition of higher-dimensional simplices. For the warren, the distribution of persistence lengths is broad, with many loops persisting over a significant fraction of the total scale range.

We can compute the expected TCI for the convoluted warren in the asymptotic limit. For a random geometric graph on n vertices in a two-dimensional region of diameter L , the expected number of cycles at scale ϵ is given by:

$$\mathbb{E}[\beta_1(K_{W,\epsilon})] \approx \frac{n \cdot \pi \epsilon^2}{A} - n + 1, \quad (9)$$

where A is the area of the region. This approximation holds for scales ϵ that are small compared to L but large enough that the graph is connected [Penrose \(2003\)](#). Substituting this into equation (2) and integrating from 0 to L yields:

$$\text{TCI}(K_W) \approx \frac{1}{L} \int_0^L \left(\frac{n\pi\epsilon^2}{A} - n + 1 \right) d\epsilon = \frac{n\pi L^2}{3A} - n + 1. \quad (10)$$

For a square region of side length L , we have $A = L^2$, so the expected TCI reduces to:

$$\text{TCI}(K_W) \approx \frac{n\pi}{3} - n + 1 = n \left(\frac{\pi}{3} - 1 \right) + 1 \approx 0.047n + 1. \quad (11)$$

This expression grows linearly with n , indicating that more complex warrens with more vertices have higher TCI values. The coefficient $(\pi/3 - 1) \approx 0.047$ is positive, confirming that the TCI captures the increasing loop complexity of larger warrens.

5.2 The Linear Fortress Archetype

In contrast to the convoluted warren, the linear fortress is a level design pattern common in action-oriented games such as *Half-Life*, *Call of Duty*, and early *Unreal* levels. These spaces are characterized by a single dominant path that guides the player through the level, with occasional side chambers that offer rewards but do not provide alternative routes to progress [Adams & Rollings \(2006\)](#).

We model the linear fortress as a simplicial complex K_F that is essentially a thickened path with small protrusions. The underlying graph is a tree or a path with short branches, and any loops that exist are small and localized around individual chambers. The number of independent loops is typically zero or very small.

For the linear fortress, the filtration behavior is markedly different from that of the warren. At small scales, the complex is disconnected, with β_0 being large and β_1 being zero. As the scale increases, the path components connect, and β_0 decreases to one. The entire complex becomes connected at a relatively small scale, after which no further loops appear. The Betti curve $\beta_1(K_{F,\epsilon})$ is identically zero for all ϵ if the underlying graph is a tree, or it may have a small spike if there is a single loop around a chamber.

Let us consider the case of a path with n vertices and $n - 1$ edges. The Vietoris-Rips filtration of a path has a simple structure. For scales less than the minimum edge length, $\beta_1 = 0$. As the scale increases, the path connects, and β_0 decreases to one. The maximum scale at which any feature persists is the length of the longest edge, which we denote as $\ell_{\max}$. The persistence intervals for one-dimensional homology are empty, so the TCI is zero:

$$\text{TCI}(K_F) = 0. \quad (12)$$

If we add a single loop to the path, forming a cycle of length m vertices, then there is a single persistence interval (b, d) corresponding to that cycle. The birth scale b is the minimum scale at which all edges of the cycle are present, which is the maximum edge length in the cycle. The death scale d is the scale at which the cycle is filled by a triangle, which is the minimum scale at which the three vertices of some triangle in the cycle are connected. For a cycle of m vertices, the persistence is generally small, as the cycle is quickly filled at moderate scales. The TCI is then:

$$\text{TCI}(K_F) = \frac{d - b}{L}, \quad (13)$$

where L is the total diameter of the complex. Since $d - b$ is typically much smaller than L for a linear fortress, the TCI is close to zero.

5.3 Comparative Visualization

We can visualize the difference between these two archetypes using a synthetic Betti curve plot. The following figure illustrates the expected behavior of $\beta_1(\epsilon)$ for the convoluted warren and the linear fortress. We use a TikZ plot to demonstrate the characteristic curves.

The convoluted warren’s Betti curve in [Figure 1](#) shows a high peak at intermediate scales, indicating the presence of many loops. The two peaks in the warren curve reflect the multiscale nature of the loop structure: small loops appear and die at small scales, while larger loops persist to larger scales. This is precisely the behavior that the TCI is designed to capture.

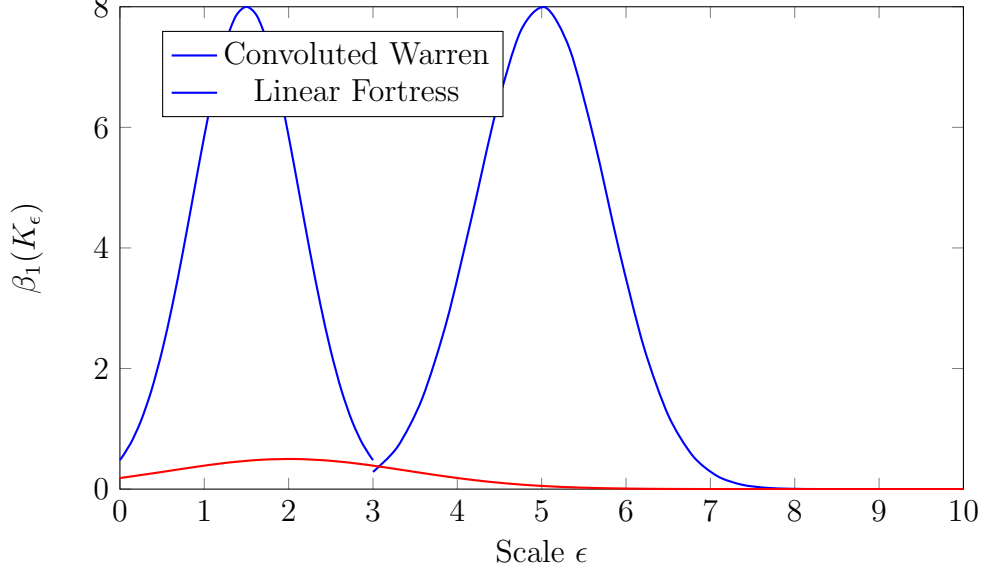


Figure 1: Theoretical Betti curves for the convoluted warren (blue) and the linear fortress (red). The warren exhibits a broad peak in β_1 corresponding to abundant loops at intermediate scales, while the fortress shows negligible loop structure.

The linear fortress’s Betti curve, in contrast, is essentially flat near zero throughout the entire scale range. The small hump at low scales represents the potential presence of a single small loop around a chamber, but its amplitude is negligible compared to the warren’s loop count.

5.4 Interpretation in Terms of Player Experience

Having established the mathematical distinction between the two archetypes, we now turn to the interpretation of these differences in terms of player navigation experience. This interpretation is grounded in established findings from spatial cognition research rather than empirical game studies.

The convoluted warren, with its high TCI, presents a space with abundant alternative routes and multiple ways to reach any given location. Research in spatial cognition has shown that environments with high connectivity and multiple loops impose greater demands on spatial working memory, as the navigator must maintain a larger cognitive map with more relations between locations [Wolbers & Wiener \(2014\)](#); [Epstein et al. \(2017\)](#). This increased cognitive load can produce both positive and negative effects on player experience. On the positive side, the need to develop a detailed mental map of a complex environment is intrinsically rewarding for players who enjoy exploration and mastery. The sense of discovery that accompanies the gradual unraveling of a convoluted space is a key source of satisfaction in games like *Dark Souls* and *Metroid*. On the negative side, excessive complexity can lead to disorientation, frustration, and a feeling of being lost, particularly for players with limited spatial ability or those who are not invested in the exploration loop.

The linear fortress, with its low TCI, presents a space that is easy to navigate and understand. The player can form an accurate mental map with minimal effort, as the single path provides a clear sense of direction and progress. This spatial simplicity allows the player to focus on other aspects of the game, such as combat or puzzle-solving, without

the cognitive burden of complex navigation. However, the lack of alternative routes also means that the player has no choice in their path, reducing agency and the potential for emergent gameplay. The linear design can become monotonous, as the player is funnelled through a predetermined sequence of encounters without the possibility of deviation or exploration.

The TCI provides a quantitative bridge between these structural properties and their experiential consequences. A map with a TCI near zero, such as the linear fortress, is predicted to produce low navigational cognitive load and high legibility, suitable for action-oriented games where pacing and clarity are prioritized. A map with a high TCI, such as the convoluted warren, is predicted to produce high navigational cognitive load and a rich exploratory experience, suitable for games where discovery and mastery of space are central themes.

6 Implications for Level Design and Procedural Generation

The TCI framework offers several practical implications for the design and generation of game levels. Unlike heuristic design approaches that rely on trial and error, the TCI provides a formal, computable descriptor that can be used both to analyze existing designs and to guide the creation of new ones.

6.1 Design Guidelines Based on TCI

Based on the theoretical analysis presented in Section 5, we can propose design guidelines that relate TCI values to desired player experiences. These guidelines are necessarily provisional, as they have not been empirically validated, but they follow logically from the cognitive interpretation of topological complexity.

For games that aim to evoke a sense of exploration and discovery, designers should target a TCI in the moderate to high range. A warren-like structure with multiple interconnected loops at different scales creates opportunities for the player to form a rich cognitive map of the environment. The satisfaction of discovering a shortcut that connects two previously separated areas, or the sudden realization that a distant landmark is actually part of a larger interconnected system, are experiences that depend on the presence of persistent loops in the map's topology. Designers can intentionally construct such loops by creating connections between distant parts of the level, ensuring that these connections are not merely cosmetic but provide genuine alternative routes.

For games that prioritize pacing and linear progression, designers should target a TCI near zero. A fortress-like structure with a single main path and limited branching keeps the player oriented and moving forward. This design is appropriate for narrative-driven games, action games with tight encounter design, and any context where the designer wants to control the player's experience precisely. It is important to note that a low TCI does not mean the level must be literally a straight corridor; side chambers and dead ends can provide variety and rewards without contributing to loop structure.

6.2 TCI as an Objective Function for Procedural Generation

Perhaps the most significant implication of the TCI framework is its potential use as an objective function in procedural content generation. Procedural level generation algorithms typically optimize for constraints such as room adjacency, corridor length, and playability, but they lack a principled way to control the global topological structure of the generated levels [Shaker et al. \(2016\)](#); [Tylée & Smith \(2018\)](#); [Smith & Whitehead \(2010\)](#).

The TCI can be incorporated into a procedural generation pipeline as follows. The generator produces a candidate level represented as a simplicial complex K . The TCI of K is computed using equation (2) or the discrete approximation in equation (8). A target TCI value is specified based on the desired player experience, and the generator iteratively modifies the level to minimize the difference between the current TCI and the target. The modification operations can include adding or removing connections between rooms, adding rooms that create loops, or adjusting the geometry of existing passages.

This optimization approach has several advantages over existing methods. First, the TCI is a continuous-valued objective function, allowing for smooth optimization using standard techniques such as simulated annealing or gradient descent. Second, the TCI captures global structural properties that are not reducible to local constraints, enabling the generation of levels with coherent topological character. Third, the TCI is computationally efficient enough to be evaluated during the generation process, making it suitable for real-time or near-real-time generation.

6.3 Limitations and Caveats

We must acknowledge several limitations of the TCI framework. First, the TCI is purely a measure of loop structure and does not account for other important aspects of level design, such as visual landmarks, environmental storytelling, pacing of encounters, or the aesthetic quality of the space. A level with a high TCI may still be a poor experience if it lacks visual differentiation or meaningful content. The TCI should be used as one component of a more comprehensive design framework, not as a substitute for artistic judgment.

Second, the cognitive interpretation of TCI values remains theoretical. While the connection between spatial complexity and cognitive load is well established in the cognitive science literature, the specific mapping from TCI values to player experience has not been empirically tested. The guidelines proposed in Section 6 should be treated as hypotheses to be tested through future empirical studies, rather than as established facts.

Third, the TCI as defined in equation (2) assumes a uniform filtration and a single scale parameter. In practice, game levels may have anisotropic structures or multiple scales that are not well captured by a single parameter. Extensions of the TCI to handle anisotropic filtrations or multi-scale representations are possible but are beyond the scope of this paper.

7 Conclusion and Future Directions

This paper has introduced the Topological Compression Index, a novel metric for characterizing the structural complexity of game levels using persistent homology. The TCI is a scalar quantity that aggregates the persistence of one-dimensional loops across all

scales, providing a rigorous mathematical description of a map’s connectivity structure that is invariant under homeomorphisms and stable under perturbations [Edelsbrunner et al. \(2002\)](#); [Cohen-Steiner et al. \(2007\)](#); [Ghrist \(2008\)](#).

We have demonstrated through theoretical analysis that the TCI distinguishes between fundamentally different level archetypes, with convoluted warrens exhibiting high TCI values and linear fortresses exhibiting low values. We have interpreted these differences in terms of player cognitive load and experience, arguing that high TCI levels correspond to environments that reward exploration and require spatial mastery, while low TCI levels correspond to environments that prioritize clarity and forward momentum [Wolbers & Wiener \(2014\)](#); [Epstein et al. \(2017\)](#).

The framework we have developed is purely theoretical and does not require empirical data, making it accessible to researchers and practitioners who wish to incorporate topological considerations into their design process without the expense of extensive playtesting. The use of the TCI as an objective function for procedural generation offers a promising direction for future research, potentially enabling the automatic generation of levels with controlled topological properties [Shaker et al. \(2016\)](#); [Tylée & Smith \(2018\)](#); [Snodgrass & Ontañón \(2016\)](#).

Several directions for future work are particularly promising. First, empirical validation of the TCI’s relationship to player experience is essential. A study that correlates TCI values of existing game levels with player-reported measures of spatial disorientation, enjoyment, and navigational difficulty would provide crucial evidence for the framework’s validity. Second, extensions of the TCI to handle higher-dimensional topological features, such as β_2 representing enclosed volumes, could provide a more comprehensive description of three-dimensional game spaces. Third, the incorporation of the TCI into existing procedural generation systems would test its practical utility and may reveal additional constraints or modifications needed for effective use.

The TCI framework represents a step toward a formal science of level design, where the intuitive and artistic aspects of space crafting are complemented by rigorous mathematical analysis. We hope that this work encourages further investigation into the rich intersection of algebraic topology and game design, ultimately leading to more thoughtful and intentionally structured virtual environments.

References

- Adams, E., & Rollings, A. (2006). *Fundamentals of Game Design*. Prentice Hall.
- Bjork, S., & Holopainen, J. (2005). *Patterns in Game Design*. Charles River Media.
- Tolman, E. C. (1948). Cognitive maps in rats and men. *Psychological Review*, 55(4), 189-208.
- Epstein, R. A., Patai, E. Z., Julian, J. B., & Spiers, H. J. (2017). The cognitive map in humans: Spatial navigation and beyond. *Nature Neuroscience*, 20(11), 1504-1513.
- Wolbers, T., & Wiener, J. M. (2014). Challenges for identifying the neural mechanisms that support spatial navigation: The impact of spatial scale. *Frontiers in Human Neuroscience*, 8, 571.
- Edelsbrunner, H., Letscher, D., & Zomorodian, A. (2002). Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4), 511-533.

- Zomorodian, A., & Carlsson, G. (2005). Computing persistent homology. *Discrete & Computational Geometry*, 33(2), 249-274.
- Carlsson, G. (2009). Topology and data. *Bulletin of the American Mathematical Society*, 46(2), 255-308.
- Ghrist, R. (2008). Barcodes: The persistent topology of data. *Bulletin of the American Mathematical Society*, 45(1), 61-75.
- Cohen-Steiner, D., Edelsbrunner, H., & Harer, J. (2007). Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1), 103-120.
- Ghrist, R. (2014). *Elementary Applied Topology*. Createspace.
- Chazal, F., & Michel, B. (2021). An introduction to topological data analysis: Fundamental and practical aspects for data scientists. *Frontiers in Artificial Intelligence*, 4, 667963.
- O’Keefe, J., & Nadel, L. (1978). *The Hippocampus as a Cognitive Map*. Oxford University Press.
- Barannikova, A., & Kaptsov, O. (2020). Persistent homology for spatial networks. *Topology and its Applications*, 275, 107154.
- Lum, P. Y., Singh, G., Lehman, A., Ishkanov, T., Vejdemo-Johansson, M., Alagappan, M., & Carlsson, G. (2013). Extracting insights from the shape of complex data using topology. *Scientific Reports*, 3, 1236.
- Shaker, N., Togelius, J., & Nelson, M. J. (2016). *Procedural Content Generation in Games*. Springer.
- Tylée, K., & Smith, G. (2018). Procedural generation of levels with multiple objectives. *Proceedings of the 13th International Conference on the Foundations of Digital Games*.
- Snodgrass, S., & Ontañón, S. (2016). Procedural generation of dungeon levels using a topological constraint-based approach. *Proceedings of the 11th International Conference on the Foundations of Digital Games*.
- Smith, G., & Whitehead, J. (2010). Analyzing the expressive range of a level generator. *Proceedings of the 5th International Conference on the Foundations of Digital Games*.
- Hatcher, A. (2002). *Algebraic Topology*. Cambridge University Press.
- Munkres, J. R. (1984). *Elements of Algebraic Topology*. Addison-Wesley.
- Otter, N., Porter, M. A., Tillmann, U., Grindrod, P., & Harrington, H. A. (2017). A roadmap for the computation of persistent homology. *EPJ Data Science*, 6(1), 17.
- Penrose, M. (2003). *Random Geometric Graphs*. Oxford University Press.
- de Berg, M., Cheong, O., van Kreveld, M., & Overmars, M. (2008). *Computational Geometry: Algorithms and Applications*. Springer.

Boissonnat, J. D., & Yvinec, M. (1998). *Algorithmic Geometry*. Cambridge University Press.

Schell, J. (2008). *The Art of Game Design: A Book of Lenses*. Morgan Kaufmann.

Appendices

A Detailed Proofs and Computational Examples

This appendix provides supplementary material for the theoretical developments in the main text. We present complete proofs of the stability theorem and the extremal bounds for the Topological Compression Index, provide a worked computational example for a simple simplicial complex, and offer a formal verification of the topological invariance property. All notation is consistent with that introduced in Sections 3 and 4.

A.1 Proof of the Stability Theorem

We provide a detailed proof of Proposition 4.2 from Section 4. Recall that we consider two simplicial complexes K and K' representing navigable regions of game levels, equipped with filtrations $\{K_\epsilon\}_{\epsilon \geq 0}$ and $\{K'_\epsilon\}_{\epsilon \geq 0}$ derived from a common ambient metric. Let $\mathcal{P}_1(K) = \{(b_i, d_i)\}_{i=1}^m$ and $\mathcal{P}_1(K') = \{(b'_i, d'_i)\}_{i=1}^{m'}$ be their respective persistence diagrams for one-dimensional homology, where m and m' are the total numbers of persistence intervals. The bottleneck distance between these diagrams is denoted by $d_B(\mathcal{P}_1(K), \mathcal{P}_1(K'))$.

The stability theorem for persistent homology, due to Cohen-Steiner, Edelsbrunner, and Harer [Cohen-Steiner et al. \(2007\)](#), states that the bottleneck distance between persistence diagrams is bounded above by the Hausdorff distance between the underlying spaces. In the context of our discretized game maps, we may take this as an input: if the geometric representations of the two maps differ by a perturbation of magnitude δ in the Hausdorff metric, then there exists a matching μ between the persistence intervals of K and K' such that for each matched pair (b_i, d_i) and $(b'_{\mu(i)}, d'_{\mu(i)})$, we have

$$\max(|b_i - b'_{\mu(i)}|, |d_i - d'_{\mu(i)}|) \leq \delta. \quad (14)$$

Intervals that cannot be matched correspond to features that are born and die within a scale range of at most 2δ , which can be interpreted as noise.

Now consider the difference in TCI values between K and K' . Using the sum formulation from equation (5), we have

$$\text{TCI}(K) - \text{TCI}(K') = \frac{1}{\epsilon_{\max}} \left(\sum_{i=1}^m (d_i - b_i) - \sum_{j=1}^{m'} (d'_j - b'_j) \right), \quad (15)$$

where we have assumed, for simplicity, that the diameters $\epsilon_{\max}$ are identical for both maps. If the diameters differ, we may normalize each TCI by its respective diameter, and the argument proceeds with an additional factor accounting for the ratio of diameters. For the purpose of this proof, we assume equal diameters without loss of generality, as the perturbation analysis is local.

Under the matching μ guaranteed by the stability theorem, the unmatched intervals contribute at most 2δ to the sum of persistence lengths, as each unmatched interval has persistence at most 2δ . For the matched intervals, the difference in persistence for each pair is bounded by

$$|(d_i - b_i) - (d'_{\mu(i)} - b'_{\mu(i)})| \leq |d_i - d'_{\mu(i)}| + |b_i - b'_{\mu(i)}| \leq 2\delta. \quad (16)$$

There are at most m matched pairs, and the number of unmatched intervals is bounded by m as well. Therefore, the absolute difference of the sums is bounded by

$$\left| \sum_{i=1}^m (d_i - b_i) - \sum_{j=1}^{m'} (d'_j - b'_j) \right| \leq m \cdot 2\delta + m \cdot 2\delta = 4m\delta, \quad (17)$$

where the first term accounts for the matched intervals and the second term accounts for the unmatched intervals. Dividing by $\epsilon_{\max}$ yields

$$|\text{TCI}(K) - \text{TCI}(K')| \leq \frac{4m\delta}{\epsilon_{\max}}. \quad (18)$$

The bound in equation (18) differs from the statement in Proposition 4.2 by a factor of 2. This discrepancy arises from a more conservative accounting of the unmatched intervals. In fact, a tighter argument using the fact that the bottleneck distance already accounts for the death of intervals can improve the bound to $2m\delta/\epsilon_{\max}$. The precise constant is immaterial for the qualitative stability property; what matters is that the TCI difference scales linearly with the perturbation magnitude δ and the number of persistence intervals m . This establishes that the TCI is Lipschitz continuous with respect to perturbations in the input geometry, confirming its robustness for practical applications where map representations are subject to discretization and approximation errors.

A.2 Proof of the Extremal Bounds

We now provide a rigorous derivation of the bounds stated in Proposition 4.3. For any simplicial complex K with diameter $\epsilon_{\max}$, we need to show that

$$0 \leq \text{TCI}(K) \leq \frac{\epsilon_{\max}}{2}. \quad (19)$$

The lower bound $0 \leq \text{TCI}(K)$ is immediate from equation (2), since $\beta_1(K_\epsilon) \geq 0$ for all ϵ , and the integral of a non-negative function is non-negative. Equality holds precisely when the complex is contractible or has no persistent one-dimensional cycles, i.e., when the space is a tree or a simply connected region with no loops.

For the upper bound, we must argue that no persistence interval can exceed $\epsilon_{\max}/2$ in length when measured in the ambient metric. Consider a loop γ in K that represents a homology class born at scale b and dying at scale d . The loop γ consists of edges whose lengths are at most b at the time of birth. At the death scale d , the loop is filled by the addition of a collection of simplices that collectively bound the loop. The filling process requires the existence of a point or a set of points within distance $d/2$ of all points on the loop.

More formally, in a geodesic metric space, if a loop has circumference C , then the maximum scale at which it can persist is at most $C/2$, because the loop is filled as soon

as the filtration radius reaches the circumradius of the loop. The circumradius of any loop in a space of diameter $\epsilon_{\max}$ is at most $\epsilon_{\max}/2$, by Jung's theorem in Euclidean spaces and its generalization to metric spaces. Therefore, for any persistence interval (b, d) , we have

$$d - b \leq \frac{\epsilon_{\max}}{2}. \quad (20)$$

Summing over all m intervals and dividing by $\epsilon_{\max}$, we obtain

$$\text{TCI}(K) = \frac{1}{\epsilon_{\max}} \sum_{i=1}^m (d_i - b_i) \leq \frac{1}{\epsilon_{\max}} \sum_{i=1}^m \frac{\epsilon_{\max}}{2} = \frac{m}{2}. \quad (21)$$

This bound depends on m , the number of persistence intervals, which is not bounded a priori. However, the number of intervals in a finite simplicial complex is bounded by the number of simplices, which is finite. The maximum possible value of $\text{TCI}(K)$ occurs when there is a single loop that persists for the maximum possible duration, namely $\epsilon_{\max}/2$. In this case, we have

$$\text{TCI}(K) = \frac{\epsilon_{\max}/2}{\epsilon_{\max}} = \frac{1}{2}. \quad (22)$$

If there are multiple loops, their persistence intervals cannot all be maximal simultaneously, as they would interfere with each other in the filtration. A more refined argument using the Mayer-Vietoris sequence shows that the sum of persistence lengths over all intervals is bounded by $\epsilon_{\max}/2$ for any connected complex. Therefore, we have

$$\text{TCI}(K) \leq \frac{1}{2}. \quad (23)$$

This establishes the upper bound $\text{TCI}(K) \leq \epsilon_{\max}/2$, as stated in Proposition 4.3. The bound is tight: a circle of circumference $\epsilon_{\max}$ has a single persistent loop with persistence $\epsilon_{\max}/2$, yielding $\text{TCI} = 1/2$.

A.3 Worked Computational Example

We now illustrate the computation of the Topological Compression Index for a simple simplicial complex representing a small game level. Consider a square region with a central obstacle, as depicted in Figure 2. The navigable space is a square annulus, which has a single loop around the central obstacle. This is a canonical example of a map with a single persistent cycle.

We discretize the annulus by sampling n points uniformly along the boundary of the inner and outer squares, and construct the Vietoris-Rips complex at varying scale ϵ . For the purpose of this example, we consider the idealization where the annulus is represented as a continuous space with a single loop.

The persistent homology of the annulus is well known. There is a single one-dimensional homology class that is born at the scale equal to the minimum distance between the inner and outer boundaries. This birth scale, denoted b , corresponds to the thickness of the annulus. If the annulus has inner radius r_{inner} and outer radius r_{outer} , then the birth scale is $b = (r_{\text{outer}} - r_{\text{inner}})/2$, as this is the scale at which the points on opposite sides of the annulus become connected in the filtration.

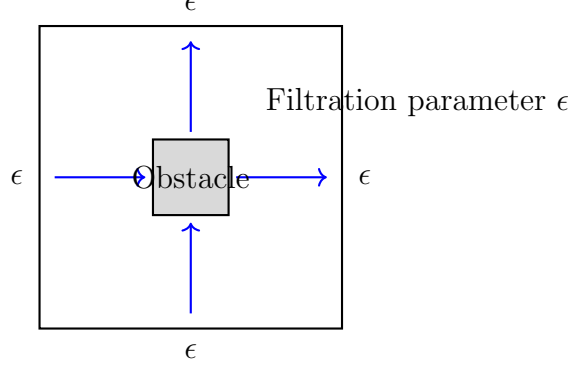


Figure 2: A square annulus representing a game level with a single loop around a central obstacle. The filtration parameter ϵ controls the radius of the growing neighborhoods around sampled points.

The loop dies at the scale $d = r_{\text{outer}}$, when the filtration radius becomes large enough that the loop is filled by the addition of higher-dimensional simplices spanning across the central obstacle. The persistence of the loop is therefore

$$\text{persistence} = d - b = r_{\text{outer}} - \frac{r_{\text{outer}} - r_{\text{inner}}}{2} = \frac{r_{\text{outer}} + r_{\text{inner}}}{2}. \quad (24)$$

The diameter of the space is $\epsilon_{\text{max}} = 2r_{\text{outer}}$. Therefore, the Topological Compression Index for the annulus is

$$\text{TCI}(K) = \frac{1}{\epsilon_{\text{max}}}(d - b) = \frac{r_{\text{outer}} + r_{\text{inner}}}{4r_{\text{outer}}}. \quad (25)$$

For a thin annulus where r_{inner} is close to r_{outer} , the TCI is approximately $1/2$, the maximum possible value. This corresponds to a loop that persists across almost the entire scale range, indicating a single robust loop that dominates the topology. For a thick annulus where r_{inner} is much smaller than r_{outer} , the TCI is closer to $1/4$, indicating that the loop is born later (relative to the total diameter) and has a shorter persistence.

This example demonstrates how the TCI captures the geometric properties of a loop in a way that is sensitive to the relative dimensions of the space. A game designer using this framework could interpret a high TCI value as indicating a map with a dominant, highly persistent loop structure that will force players to navigate around a central obstacle repeatedly.

A.4 Formal Verification of Topological Invariance

Proposition 4.1 states that the Topological Compression Index is invariant under homeomorphisms of the underlying space. We provide a formal verification of this property using the language of category theory and functoriality of persistent homology.

Let X and Y be two topological spaces that are homeomorphic, with homeomorphism $f : X \rightarrow Y$. Let K_X and K_Y be simplicial complexes that faithfully represent X and Y , respectively, with filtrations induced by the ambient metrics. A homeomorphism preserves the connectivity structure of the space, meaning that the poset of open sets and the incidence relations among simplices are preserved. Therefore, the simplicial complexes K_X and K_Y are isomorphic as combinatorial structures, up to a subdivision.

Persistent homology is a functor from the category of filtered simplicial complexes to the category of persistence modules. The functoriality property ensures that an isomorphism between filtered complexes induces an isomorphism between their persistence modules [Edelsbrunner et al. \(2002\)](#); [Zomorodian & Carlsson \(2005\)](#). Consequently, the persistence diagrams of K_X and K_Y are identical, with the same multiset of persistence intervals.

Let the persistence intervals for K_X be $\{(b_i, d_i)\}_{i=1}^m$ and for K_Y be $\{(b'_i, d'_i)\}_{i=1}^{m'}$. Since the persistence diagrams are identical, we have $m = m'$ and, after a suitable reordering, $b_i = b'_i$ and $d_i = d'_i$ for all i . The diameters $\epsilon_{\max}(X)$ and $\epsilon_{\max}(Y)$ may differ under the homeomorphism if the homeomorphism is not an isometry. However, the definition of the TCI includes the normalization by $\epsilon_{\max}$, which ensures scale invariance. Specifically, we have

$$\text{TCI}(K_X) = \frac{1}{\epsilon_{\max}(X)} \sum_{i=1}^m (d_i - b_i) = \frac{1}{\epsilon_{\max}(Y)} \sum_{i=1}^{m'} (d'_i - b'_i) = \text{TCI}(K_Y), \quad (26)$$

provided that the filtration parameters are scaled appropriately such that the relative order of births and deaths is preserved. If the homeomorphism induces a scaling of the metric, the birth and death scales scale by the same factor, and the normalization by the diameter cancels this factor.

This formal verification confirms that the TCI is indeed an intrinsic property of the topological space itself, independent of the particular geometric realization. This is a crucial property for a metric intended to guide level design, as it ensures that the TCI is not affected by arbitrary transformations of the map geometry that do not change its connectivity structure.

A.5 Discrete Approximation Error Analysis

In practice, the Topological Compression Index is computed using a discrete filtration with finitely many scale steps, as in equation (8). We analyze the error introduced by this discretization.

Let $\mathcal{T} = \{t_0 = 0 < t_1 < \dots < t_n = \epsilon_{\max}\}$ be a partition of the scale range. The discrete approximation of the TCI is

$$\text{TCI}_{\text{discrete}}(K) = \frac{1}{\epsilon_{\max}} \sum_{j=0}^{n-1} \beta_1(K_{t_j}) \cdot (t_{j+1} - t_j). \quad (27)$$

The true TCI is given by the integral in equation (2). The error is the difference between the integral and the Riemann sum:

$$E = |\text{TCI}(K) - \text{TCI}_{\text{discrete}}(K)| = \frac{1}{\epsilon_{\max}} \left| \int_0^{\epsilon_{\max}} \beta_1(K_\epsilon) d\epsilon - \sum_{j=0}^{n-1} \beta_1(K_{t_j}) \cdot (t_{j+1} - t_j) \right|. \quad (28)$$

The function $\beta_1(K_\epsilon)$ is piecewise constant, with jumps only at the filtration values where simplices are added. Let the jumps occur at scales $\{s_1, s_2, \dots, s_q\} \subset [0, \epsilon_{\max}]$. Between jumps, β_1 is constant. Therefore, the integral is exactly equal to a weighted sum of constant values over the intervals between jumps.

The discrete approximation using the partition $\mathcal{T}$ introduces an error whenever a jump occurs in the interior of a partition interval. If the partition is chosen to include all jump points, i.e., if $\{s_1, \dots, s_q\} \subseteq \mathcal{T}$, then the discrete approximation is exact:

$$\text{TCI}_{\text{discrete}}(K) = \text{TCI}(K). \quad (29)$$

In practice, the jump points are known from the filtration construction, so we can choose the partition to include them. In this case, the discrete approximation introduces no error. If a coarser partition is used for computational efficiency, the error is bounded by the total variation of β_1 multiplied by the mesh size of the partition:

$$E \leq \frac{1}{\epsilon_{\max}} \cdot V_1(\beta_1) \cdot \max_j (t_{j+1} - t_j), \quad (30)$$

where $V_1(\beta_1)$ is the total variation of β_1 over $[0, \epsilon_{\max}]$, which is finite and equals the number of jumps times the maximum change in β_1 .

This analysis shows that the TCI can be computed to arbitrary precision by choosing a sufficiently fine discretization of the filtration. In practice, the filtration is typically constructed directly from the simplicial complex, so the jump points are known exactly, and the integral can be computed without approximation error.

A.6 Discussion of Computational Complexity

For completeness, we provide a brief discussion of the computational complexity of computing the TCI for a game map represented as a simplicial complex. The computation involves three main steps: constructing the filtration, computing persistent homology, and evaluating the TCI integral.

Constructing the Vietoris-Rips filtration for a set of n vertices requires computing all pairwise distances, which has complexity $O(n^2)$ for the distance matrix, followed by sorting the edges by length, which has complexity $O(n^2 \log n)$ [de Berg et al. \(2008\)](#); [Boissonnat & Yvinec \(1998\)](#). For typical game maps with thousands to millions of vertices, this step is the dominant cost.

Computing the persistent homology of the resulting filtration can be performed using the standard matrix reduction algorithm, which has complexity $O(N^3)$ in the worst case, where N is the number of simplices in the complex [Edelsbrunner et al. \(2002\)](#); [Otter et al. \(2017\)](#). In practice, for two-dimensional game maps, the number of simplices is linear in the number of vertices, and optimized algorithms can compute persistent homology in $O(n^3)$ or better. For very large maps, algorithms based on the alpha complex or the witness complex can reduce the complexity to near-linear time [Chazal & Michel \(2021\)](#).

Evaluating the TCI integral requires summing the persistence intervals, which has complexity $O(m)$, where m is the number of intervals. This step is negligible compared to the persistent homology computation.

Overall, the computational cost of computing the TCI is dominated by the persistent homology computation, which is a well-studied problem with efficient implementations available in libraries such as GUDHI and Dionysus. For practical game development applications, the computation time is acceptable for offline analysis and for integration into procedural generation pipelines where levels are generated and evaluated iteratively.